# Refugee Inflows and Native Parity Progression

## Evidence from Turkey

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## Motivation

1. Impact of immigration on native fertility is studied, but **not in the context of refugees**.
2. Fertility rates in OECD countries have fallen by half over the past 60 years to roughly **1.5 children per woman**.
3. **Turkey: a unique case study**
   - Host to one of the world's **largest refugee populations** ($\approx$5.3M Syrians registered, $\approx$3.6M in Turkey).
   - Native total fertility now **below the OECD average** ($\approx$1.41).
   - Ideal setting for studying how migration shocks interact with **low fertility**.

**Research question.** How does the influx of Syrian refugees affect the fertility decisions of native Turkish women?

We look at both **marriage** outcomes and **child-parity** outcomes (one to five children) at economically meaningful ages.

# Preview — What We Find and How We Read It

**Main Findings.**

- **No effect on marriage or early-parity fertility** (parities 1–3, all ages).
- Raw 2SLS: large positive estimates at **higher parities (4–5)** in the **late 20s / 30s**.
- Response **concentrates in three subgroups**: Arabic/Kurdish mother tongue, lower-education, and high-exposure East/Southeast border region.

**Two potential mechanisms.**

- **Labor-market / opportunity-cost** (Aksu, Erzan, Kırdar 2022): refugee inflow reduced *native female* employment but not male employment $\Rightarrow$ lower opportunity cost of an extra child without an income hit.
- **Ideal-fertility / norms**: stated ideal-$n$ shifts upward at the *same upper thresholds* and in the *same subgroups* where realised higher-parity fertility responds.

# Data, Outcomes, and Exposure

**TDHS:** nationally representative microdata on women with rich **fertility histories** and **migration histories** for responding women.

**Main data:** TDHS **2013 & 2018**; woman×age **pseudo-panel** from age 12 to survey age; province assigned by **2011 residence** (pre-shock).

**Cumulative parity outcomes** (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \qquad p \in \{1,2,3,4,5\}, \quad a \in \{16,20,24,28,32,36\}.$$

Plus the marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

**Exposure: average refugee-to-native ratio over the woman's age window 12 to $a$:**

$$\bar{R}_{ja} = \frac{1}{a-11} \sum_{s=12}^{a} R_{j,\, t(s,i)}.$$

**Interpretation.** A 1 pp. higher *average annual* ratio over ages 12–$a$ raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

Sample details Pseudo-panel Data sources

# Identification — Quasi-Exogenous Distance Variation

- **Challenge.** Refugee location is endogenous: internal migration responds to local labor-market conditions; by 2018 refugees had spread well beyond border provinces.
- **Strategy.** Follow Aksu, Erzan, Kırdar (2022): instrument $\bar{R}_{ja}$ with a shift-share variable based on geographic distance from Syrian to Turkish provinces.

$$Z_{n,t} = \underbrace{\sum_{s=1}^{13} \frac{(1/d_{s,T})\,\pi_s}{\sum_{X\in\{T,L,J,I\}}(1/d_{s,X})}}_{\text{share weight (time-invariant)}} \cdot T_t = B_n \times T_t.$$

$\pi_s$: pre-war pop. share of Syrian province $s$; $d_{n,s}$: distance Syrian $s$ to Turkish $n$; $T_t$: aggregate Syrian refugee stock across four host countries.

**Where the variation comes from.**

- $B_n$ is a **time-invariant, geography-based** intensity of refugee exposure for province $n$: provinces closer to Syrian provinces have higher $B_n$.
- $T_t$ scales it over time as the aggregate Syrian refugee stock grows post-2011, so the variation that identifies $\beta$ is $B_n \times$ (post-shock years) — a **continuous-treatment difference-in-differences** design.
- **Exclusion restriction:** conditional on FE and trend controls, $B_n$ is **uncorrelated with pre-shock fertility trends** among Turkish natives.

$T_t$ over time

# Settlement-Propensity Weight $B_n$ across Provinces

Settlement-propensity weight $B_n$ across Turkish provinces (quintiles)

# Empirical Methodology — Estimating Equation

For each woman $i$ in province $j$, we model whether parity $p$ has been reached by age $a$:

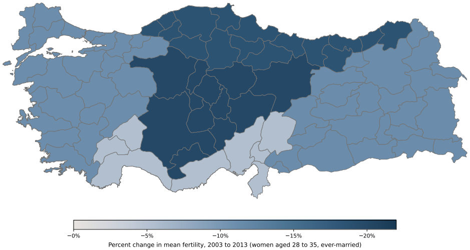
$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)} \bar{R}_{ja} + \mathrm{X}_i' \Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a,i) + \lambda_{l(i)} \cdot t(a,i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

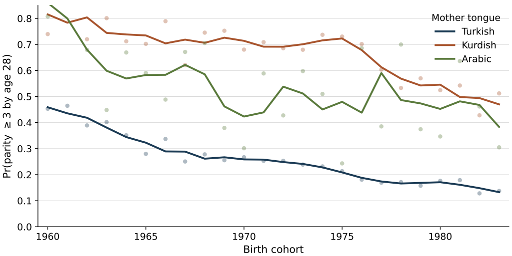
- $\beta^{(p,a)}$ — coefficient of interest: effect of a 1 pp. higher *average annual* refugee-to-native ratio on Pr(parity $\geq p$ by age $a$).
- $\mathrm{X}_i'$ — individual controls: survey wave, childhood urban/rural, mother's education, mother tongue.
- $\delta_j$ — province FE; $\mu_{t(a,i)}$ — year-at-age-$a$ FE ($\equiv$ birth-cohort FE at fixed $a$).
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear trends.
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear trends (Turkish / Kurdish / Arabic).
- Trends start in 1987; SEs clustered at province; DHS sampling weights.

# Why Region-Specific and Mother-Tongue-Specific Trends?

## Region trends




## Mother-tongue trends



# Placebo — $Y \sim B_n \times t$ on Pre-Shock Cohorts

On the pre-shock subsample (cohort-age window ends before 2012), replace the shift-share instrument $\bar{Z}_{ja}$ with $B_j \times t(a,i)$ — a linear placebo interaction in calendar time. Under the null of no pre-trends in $B_j$, $\beta(p,a)$ should be zero.

$$Y_{ija}^{(p)} = \alpha + \gamma^{(p,a)}\left(B_j \times t(a,i)\right) + X_i'\Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)}t + \varepsilon_{ija}.$$

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | 0.0003 | 0.0010 | 0.0012* | 0.0008** | 0.0011* | -0.0001 |
| | (0.0005) | (0.0007) | (0.0007) | (0.0004) | (0.0006) | (0.0007) |
| | [0.84] | [0.47] | [0.47] | [0.38] | [0.47] | [0.90] |
| Ever Gave Birth to One Child | 0.0005* | 0.0011 | 0.0007 | 0.0011** | 0.0011** | 0.0001 |
| | (0.0003) | (0.0008) | (0.0010) | (0.0005) | (0.0005) | (0.0007) |
| | [0.77] | [0.92] | [1.00] | [0.62] | [0.74] | [1.00] |
| Ever Gave Birth to Two Children | 0.0001 | -0.0001 | 0.0009 | 0.0018*** | 0.0015 | 0.0006 |
| | (0.0002) | (0.0004) | (0.0006) | (0.0007) | (0.0010) | (0.0011) |
| | [1.00] | [1.00] | [0.91] | [0.48] | [0.91] | [1.00] |
| Ever Gave Birth to Three Children | 0.0000 | -0.0000 | 0.0004 | 0.0000 | -0.0007 | -0.0024* |
| | (0.0000) | (0.0002) | (0.0007) | (0.0008) | (0.0008) | (0.0013) |
| | [0.92] | [1.00] | [1.00] | [1.00] | [0.99] | [0.79] |
| Ever Gave Birth to Four Children | – | – | 0.0002 | 0.0011 | 0.0005 | -0.0005 |
| | (–) | (–) | (0.0004) | (0.0009) | (0.0023) | (0.0022) |
| | | | [1.00] | [0.92] | [1.00] | [1.00] |
| Ever Gave Birth to Five Children | – | – | 0.0000 | 0.0002 | 0.0012 | 0.0018 |
| | (–) | (–) | (0.0002) | (0.0005) | (0.0014) | (0.0022) |
| | | | [1.00] | [1.00] | [0.99] | [0.99] |
| Observations | 12,355 | 11,560 | 10,226 | 8,194 | 6,063 | 4,098 |

Near-zero $\hat{\gamma}^{(p,a)}$. Stars = raw inference; **[brackets]** = Romano–Wolf stepdown $p$-values across the 32-cell family. Under **Romano–Wolf**, **no cell rejects**.

What is RW? First stage NUTS-1 placebo NUTS-2 placebo

# Main Results — Primary Sample, Romano–Wolf $p$-values

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | -0.047 | -0.572 | 0.513 | -0.013 | 0.220 | -0.299 |
| | (0.217) | (0.600) | (1.689) | (1.047) | (0.920) | (1.241) |
| | [1.00] | [0.90] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 1$ | 0.010 | 0.123 | -0.043 | 0.923 | -1.515 | 0.184 |
| | (0.127) | (0.523) | (1.968) | (1.342) | (1.789) | (1.371) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [0.99] | [1.00] |
| Ever Birth $\geq 2$ | 0.028 | 0.305 | -0.151 | 0.613 | 1.740 | -0.845 |
| | (0.041) | (0.546) | (0.746) | (1.906) | (3.419) | (1.189) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 3$ | 0.008 | 0.382 | -0.371 | 1.250 | 2.893 | 1.062 |
| | (0.005) | (0.248) | (0.565) | (1.559) | (2.700) | (1.810) |
| | [0.88] | [0.88] | [1.00] | [1.00] | [0.98] | [1.00] |
| Ever Birth $\geq 4$ | – | – | 0.250 | 1.905** | 2.246 | 4.252** |
| | (–) | (–) | (0.340) | (0.809) | (1.415) | (1.963) |
| | | | [1.00] | [0.52] | [0.86] | [0.61] |
| Ever Birth $\geq 5$ | – | – | 0.313 | 1.392* | 1.414 | 4.133** |
| | (–) | (–) | (0.232) | (0.803) | (0.912) | (1.856) |
| | | | [0.92] | [0.81] | [0.88] | [0.58] |
| Observations | 14,354 | 13,484 | 12,241 | 10,240 | 8,245 | 6,110 |

Stars = raw inference; **[brackets]** = Romano–Wolf stepdown $p$-values across the 32-cell family. Positive raw effects at parities 4–5, ages 28+. **No cell survives RW** in the pooled sample.

What is RW? 4-wave EM OLS Robustness

# Heterogeneity — Lower-Educated Women (RW $p$-values)

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Lower education (no schooling / primary)* | | | | | | |
| Ever Married | -0.185 | -0.872 | 2.048 | -0.059 | 0.189 | -0.203 |
| | (0.303) | (1.285) | (1.824) | (1.529) | (1.035) | (1.190) |
| | [0.95] | [0.95] | [0.86] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 1$ | -0.023 | 0.048 | 0.291 | 1.303 | 0.045 | 0.558 |
| | (0.187) | (1.249) | (2.443) | (1.957) | (1.274) | (1.331) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 2$ | 0.003 | 0.261 | -1.116 | 1.691 | 1.602 | 1.986 |
| | (0.057) | (1.636) | (1.682) | (1.645) | (2.743) | (1.635) |
| | [1.00] | [1.00] | [1.00] | [0.98] | [1.00] | [0.96] |
| Ever Birth $\geq 3$ | 0.004 | 0.738 | -0.637 | 2.823 | 3.261 | 4.689*** |
| | (0.007) | (0.765) | (1.210) | (2.238) | (2.446) | (1.697) |
| | [1.00] | [0.99] | [1.00] | [0.96] | [0.94] | [0.33] |
| Ever Birth $\geq 4$ | – | – | -0.067 | 3.175** | 2.968*** | 6.658*** |
| | (–) | (–) | (0.679) | (1.294) | (1.053) | (2.370) |
| | | | [1.00] | [0.45] | [0.32] | [0.32] |
| Ever Birth $\geq 5$ | – | – | 0.351 | 1.787 | 1.893 | 5.619*** |
| | (–) | (–) | (0.414) | (1.284) | (1.160) | (1.860) |
| | | | [0.99] | [0.93] | [0.85] | [0.26] |
| Observations | 9,361 | 9,132 | 8,729 | 7,633 | 6,348 | 4,897 |

Those with primary school or less. Sharp positive raw effects at parities **4–5, ages 28–36**; bracketed RW $p$-values **rise substantially** (smallest 0.26); **no cell rejects at** $\alpha = 0.05$.

Baseline rates EM analog Higher-educated

# Heterogeneity — Kurdish / Arabic Mother Tongue (RW $p$-values)

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Arabic / Kurdish mother tongue* | | | | | | |
| Ever Married | 0.556* | 0.520 | 0.585 | -4.025* | 1.933 | 0.859 |
| | (0.323) | (1.759) | (2.478) | (2.239) | (2.238) | (1.146) |
| | [0.53] | [0.96] | [0.96] | [0.52] | [0.87] | [0.87] |
| Ever Birth $\geq 1$ | 0.336* | 1.821 | -1.754 | -1.585 | 2.441 | 0.762 |
| | (0.177) | (1.593) | (2.323) | (2.396) | (2.770) | (1.811) |
| | [0.73] | [0.95] | [0.99] | [0.99] | [0.98] | [0.99] |
| Ever Birth $\geq 2$ | 0.047 | 1.806 | 1.433 | 1.460 | 4.418 | 1.994 |
| | (0.073) | (1.166) | (1.838) | (2.987) | (3.288) | (1.401) |
| | [0.99] | [0.82] | [0.99] | [0.99] | [0.89] | [0.86] |
| Ever Birth $\geq 3$ | -0.019 | 0.647 | -0.869 | 12.305*** | 6.019* | 2.569* |
| | (0.026) | (1.229) | (1.230) | (2.469) | (3.479) | (1.522) |
| | [0.99] | [0.99] | [0.99] | [0.04] | [0.78] | [0.79] |
| Ever Birth $\geq 4$ | – | – | 1.201 | 8.119*** | 2.024 | 8.244*** |
| | (–) | (–) | (1.318) | (1.752) | (3.811) | (1.788) |
| | | | [0.98] | [0.05] | [0.99] | [0.05] |
| Ever Birth $\geq 5$ | – | – | 0.394 | 3.155 | 4.719 | 5.038** |
| | (–) | (–) | (1.017) | (2.450) | (3.347) | (2.366) |
| | | | [0.99] | [0.92] | [0.86] | [0.64] |
| Observations | 3,145 | 2,785 | 2,415 | 1,951 | 1,489 | 1,055 |

**One subgroup that survives Romano–Wolf**: parity 3@28 ($p^{\mathrm{RW}} = 0.04$), parity 4@28 ($p^{\mathrm{RW}} = 0.05$), parity 4@36 ($p^{\mathrm{RW}} = 0.05$). Turkish-speaking women null.

Baseline rates EM analog Turkish MT

# Heterogeneity — High-Exposure Border Region (RW $p$-values)

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *High-exposure border region (East / Southeast)* | | | | | | |
| Ever Married | -0.245 | -1.691*** | -2.591 | -0.913 | -0.430 | 1.565* |
| | (0.358) | (0.646) | (1.976) | (1.656) | (0.761) | (0.855) |
| | [0.87] | [0.27] | [0.67] | [0.87] | [0.87] | [0.50] |
| Ever Birth $\geq 1$ | -0.023 | -0.404 | -3.030* | 1.446 | 0.339 | 1.989* |
| | (0.154) | (0.478) | (1.569) | (2.261) | (1.428) | (1.185) |
| | [1.00] | [0.98] | [0.65] | [0.99] | [1.00] | [0.70] |
| Ever Birth $\geq 2$ | 0.021 | 1.425*** | 1.283 | 0.470 | 5.495** | 0.330 |
| | (0.052) | (0.471) | (0.934) | (2.126) | (2.469) | (1.105) |
| | [1.00] | [0.32] | [0.83] | [1.00] | [0.56] | [1.00] |
| Ever Birth $\geq 3$ | -0.015 | 0.339 | -0.050 | 5.112** | 9.121*** | 4.400** |
| | (0.019) | (0.429) | (0.599) | (2.045) | (2.816) | (1.763) |
| | [0.98] | [0.98] | [1.00] | [0.49] | [0.29] | [0.49] |
| Ever Birth $\geq 4$ | – | – | 0.229 | 2.987** | 6.751*** | 5.753** |
| | (–) | (–) | (0.575) | (1.283) | (1.698) | (2.491) |
| | | | [1.00] | [0.55] | [0.15] | [0.55] |
| Ever Birth $\geq 5$ | – | – | -0.014 | 1.875** | 4.148*** | 6.594* |
| | (–) | (–) | (0.427) | (0.934) | (1.390) | (3.499) |
| | | | [1.00] | [0.62] | [0.33] | [0.65] |
| Observations | 4,002 | 3,522 | 3,022 | 2,404 | 1,851 | 1,311 |

East / Southeast (highest $B_n$, $\sim$55% Kurdish-speaking). Largest raw point estimates at parities **3–5, ages 24+**; RW $p$-values rise but cells do not reject at $\alpha = 0.05$. Rest-of-Turkey null.

Baseline rates | EM analog | Rest of Turkey

# Mechanism I — Labor-Market / Opportunity-Cost Channel

For $Y_i \in \{W_i, H_i\}$ (woman currently employed; husband worked last 7 days), on the TDHS women whose fertility we study, with $\bar{R}$ instrumented by $B_{j(i)}\,\bar{T}_{a_i^*}$:

$$Y_i \;=\; \beta\,\bar{R}_{j(i),a_i^*} + \alpha_{j(i)} + \mu_{t_i} + X_i'\gamma + \sum_r \lambda_r\,D_{it}^r + \varepsilon_i.$$

| Subgroup | (1) Woman employed | (2) Husband worked 7d |
|---|---|---|
| Pooled | $-2.7^{**}$ | $+2.0^{*}$ |
| Lower education | $-4.8^{***}$ | $+2.8^{**}$ |
| Arabic / Kurdish mother tongue | $-3.9^{**}$ | $\approx 0$ |
| High-exposure border region (E/SE) | $-2.2^{*}$ | $\approx 0$ |

**Gender-asymmetric pattern:** female employment falls in the pooled sample and in every subgroup where the higher-parity fertility response is concentrated; husband employment is stable or higher — same signature as Aksu, Erzan, Kırdar (2022).

# Mechanism II — Ideal Number of Children, by Subgroup

Cross-sectional 2SLS of stated ideal $n_i^\star$ (and threshold cuts $\mathbb{1}\{n_i^\star \geq k\}$) on $\bar{R}_{j(i),t(i)}$, ever-married four-wave sample. Same controls / FE / trends as the fertility specification.

| Subgroup | Primary | Threshold decomposition | | | |
|---|---|---|---|---|---|
| | Ideal $n$ | Ideal $\geq 2$ | Ideal $\geq 3$ | Ideal $\geq 4$ | Ideal $\geq 5$ |
| Full sample (pooled) | 1.197** | 0.046 | 0.174 | 0.514* | 0.256** |
| | (0.541) | (0.061) | (0.254) | (0.270) | (0.109) |
| Lower education | 1.533*** | 0.001 | 0.203 | 0.676*** | 0.351*** |
| | (0.549) | (0.088) | (0.247) | (0.247) | (0.101) |
| Arabic / Kurdish mother tongue | 3.034** | 0.129 | 0.512 | 0.911*** | 0.475*** |
| | (1.253) | (0.158) | (0.417) | (0.350) | (0.184) |
| High-exposure border region | 1.566** | 0.069 | 0.212 | 0.727** | 0.368** |
| | (0.734) | (0.117) | (0.237) | (0.336) | (0.143) |

**Stated ideal *n* shifts up at the upper thresholds** ($\geq 4$, $\geq 5$), strongest in the three treated subgroups (low-educ, Arabic/Kurdish, high-exposure border region). A shift in *desired* family size is hard to read as pure labor-market response: complementary **norms / preference** channel.

# Conclusion

**The Syrian inflow raised native parity progression in Kurdish/Arabic, lower-educated, and southeast subgroups.**

- **No effect** on marriage or first births anywhere.
- Pooled higher-parity (4–5) estimates are large and positive under raw inference but **do not survive Romano–Wolf** across 32 tests.
- Effects **concentrate** in three overlapping groups: **Kurdish/Arabic mother tongue**, **lower-educated**, **heavily-treated southeast**. Arabic/Kurdish subgroup survives RW.
- Pattern (parity-progression, not entry into motherhood) is most consistent with a **gender-asymmetric labor-market** channel; ideal-$n$ shifts at upper thresholds add a complementary norms reading.

**Thank you. / Questions?**

AAO comparison

# Appendix

# Appendix: Romano–Wolf Multiple-Testing Correction

**The problem.** The main fertility table is a family of **32 hypotheses** (5 parities $\times$ 6 ages $+$ 2 marriage cells). At nominal $\alpha = 0.05$, $\sim 1.6$ false rejections are expected by chance even if no real effect exists.

**What RW does.** Adjusts each *p*-value so that the probability of *any* false rejection across the whole family stays below $\alpha$ – the **familywise error rate** (FWER):

$$\text{FWER} \;=\; \Pr(\text{at least one false rejection}) \;\leq\; \alpha.$$

**How.** Bootstrap the joint distribution of *t*-statistics – preserving province-level clustering and cross-cell correlation – and adjust each cell's *p*-value to the bootstrap probability that the most extreme remaining *t*-stat exceeds the observed one (**stepdown**).

**What changes in the tables.** Cells with raw $p < 0.05$ may have RW $p > 0.05$ – the bracketed values rise; this is the price of testing 32 hypotheses simultaneously.

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# Appendix • Four-wave EM sample

*Main + three subgroups, TDHS 2003/08/13/18, childhood province.*

# Appendix: Main IV — Four-Wave Ever-Married Sample (RW)

Main IV (2SLS) results, ever-married sample.

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Gave Birth to One Child | 0.313 | -1.348 | 0.986 | -0.608 | -1.397 | 1.373** |
| | (1.914) | (1.229) | (2.238) | (1.109) | (1.437) | (0.680) |
| | [1.00] | [0.98] | [1.00] | [1.00] | [0.99] | [0.74] |
| Ever Gave Birth to Two Children | 0.322* | 1.273 | 0.687 | -1.075 | 2.253 | 0.100 |
| | (0.172) | (2.240) | (1.452) | (2.018) | (2.615) | (1.927) |
| | [0.81] | [1.00] | [1.00] | [1.00] | [0.99] | [1.00] |
| Ever Gave Birth to Three Children | 0.143 | 1.482 | -0.772 | -0.900 | 0.714 | -0.561 |
| | (0.117) | (0.974) | (0.924) | (1.991) | (2.365) | (2.507) |
| | [0.96] | [0.91] | [0.99] | [1.00] | [1.00] | [1.00] |
| Ever Gave Birth to Four Children | – | – | -0.368 | 0.401 | 0.029 | 2.742 |
| | (–) | (–) | (0.518) | (0.754) | (1.582) | (2.004) |
| | | | [1.00] | [1.00] | [1.00] | [0.94] |
| Ever Gave Birth to Five Children | – | – | 0.280 | 0.789 | -0.346 | 1.567 |
| | (–) | (–) | (0.305) | (0.551) | (1.274) | (1.624) |
| | | | [0.99] | [0.92] | [1.00] | [0.99] |
| Observations | 19,281 | 21,525 | 21,782 | 19,776 | 16,442 | 12,282 |

Ever-married, TDHS 2003/08/13/18, childhood province. Trade-off: more pre-shock anchoring but (i) marriage is treatment-responsive (selection); (ii) measurement error on province.

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# Appendix: Lower-Educated Women — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Lower-educated women ($\leq$ primary)* | | | | | | |
| Ever Birth $\geq 1$ | -0.452 | -1.843 | -0.507 | 0.527 | 0.251 | 1.277 |
| | (2.179) | (1.977) | (1.713) | (1.357) | (0.958) | (0.888) |
| | [1.00] | [0.99] | [1.00] | [1.00] | [1.00] | [0.91] |
| Ever Birth $\geq 2$ | 0.145 | 0.151 | -1.999 | 0.419 | 1.410 | 1.543 |
| | (0.237) | (2.977) | (2.432) | (1.279) | (2.445) | (1.342) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] | [0.97] |
| Ever Birth $\geq 3$ | 0.002 | 1.480 | -1.751 | 0.858 | 1.391 | 1.931 |
| | (0.154) | (1.370) | (1.414) | (2.529) | (1.985) | (2.720) |
| | [1.00] | [0.97] | [0.97] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 4$ | – | – | -0.605 | 1.699 | 1.063 | 4.051 |
| | (–) | (–) | (0.858) | (1.219) | (1.205) | (2.468) |
| | | | [1.00] | [0.95] | [0.99] | [0.84] |
| Ever Birth $\geq 5$ | – | – | 0.350 | 1.148 | 0.151 | 2.733* |
| | (–) | (–) | (0.412) | (0.882) | (1.468) | (1.749) |
| | | | [0.99] | [0.96] | [1.00] | [0.84] |
| Observations | 14,711 | 16,562 | 16,952 | 15,800 | 13,465 | 10,305 |

Positive higher-parity raw estimates at ages 28+ persist; no cell survives RW in the EM family.

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Main IV (2SLS) results, ever-married sample.

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Arabic / Kurdish mother tongue* | | | | | | |
| Ever Birth $\geq 1$ | 1.473 | -1.001 | -1.113 | -2.204 | 1.714 | 0.322 |
| | (1.219) | (2.029) | (2.688) | (2.129) | (1.892) | (1.488) |
| | [0.92] | [1.00] | [1.00] | [0.95] | [0.96] | [1.00] |
| Ever Birth $\geq 2$ | 0.251 | 2.834* | 2.140 | 0.810 | 2.998 | 2.097* |
| | (0.412) | (1.771) | (3.603) | (2.459) | (2.431) | (1.582) |
| | [1.00] | [0.55] | [1.00] | [1.00] | [0.90] | [0.90] |
| Ever Birth $\geq 3$ | -0.082 | 2.023 | -1.391 | 12.349*** | 3.253 | 1.314 |
| | (0.329) | (1.993) | (1.874) | (2.473) | (3.036) | (2.216) |
| | [1.00] | [0.92] | [1.00] | [0.05] | [0.96] | [1.00] |
| Ever Birth $\geq 4$ | – | – | -0.184 | 8.438** | 2.087 | 7.109*** |
| | (–) | (–) | (1.792) | (2.470) | (3.854) | (1.750) |
| | | | [1.00] | [0.43] | [1.00] | [0.17] |
| Ever Birth $\geq 5$ | – | – | 0.306 | 4.181 | 6.160* | 6.020** |
| | (–) | (–) | (0.985) | (2.658) | (2.870) | (2.091) |
| | | | [1.00] | [0.90] | [0.75] | [0.32] |
| Observations | 4,492 | 4,765 | 4,517 | 3,885 | 3,041 | 2,157 |

The parity $\geq$ 3@28 cell rejects at $\alpha = 0.05$ even in the EM sample ($p^{\mathrm{RW}} \approx 0.05$). Higher-parity raw estimates at 4@28, 4@36, 5@28 are large but do not survive RW in the EM family.

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# Appendix: High-Exposure Border Region — Four-Wave EM Sample

Main IV (2SLS) results, ever-married sample.

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *High-exposure border region (E/SE)* | | | | | | |
| Ever Birth $\geq 1$ | 3.464** | -0.142 | -0.700 | 1.105 | 1.292 | 0.781 |
| | (1.693) | (1.035) | (0.997) | (1.828) | (1.170) | (0.982) |
| | [0.36] | [1.00] | [0.97] | [0.96] | [0.85] | [0.92] |
| Ever Birth $\geq 2$ | 0.723 | 4.920*** | 3.907*** | -0.081 | 6.482*** | 0.783 |
| | (0.312) | (2.204) | (2.012) | (1.860) | (2.197) | (1.061) |
| | [0.72] | [0.17] | [0.39] | [1.00] | [0.17] | [0.96] |
| Ever Birth $\geq 3$ | 0.226 | 1.177 | -0.264 | 4.832* | 9.565*** | 4.237* |
| | (0.182) | (1.303) | (1.378) | (2.055) | (2.461) | (1.606) |
| | [0.89] | [0.88] | [1.00] | [0.54] | [0.06] | [0.60] |
| Ever Birth $\geq 4$ | – | – | 0.568 | 3.187 | 7.355*** | 8.343 |
| | (–) | (–) | (1.068) | (1.430) | (1.890) | (2.308) |
| | | | [0.97] | [0.66] | [0.10] | [0.60] |
| Ever Birth $\geq 5$ | – | – | 0.331 | 2.436** | 4.365 | 7.321** |
| | (–) | (–) | (0.587) | (1.102) | (1.601) | (3.818) |
| | | | [0.97] | [0.54] | [0.64] | [0.31] |
| Observations | 5,666 | 6,103 | 5,823 | 4,995 | 3,954 | 2,796 |

Raw higher-parity point estimates at ages 28+ remain large and positive, but **no EM cell survives RW at $\alpha = 0.05$**.

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# Appendix • Baseline parity-attainment by subgroup

*Share of women in each age column who have reached the row outcome.*

# Appendix: Baseline Parity-Attainment — Lower-Educated Women

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | 16.1% | 61.5% | 85.5% | 93.5% | 96.0% | 97.4% |
| Ever Gave Birth to One Child | 5.4% | 44.7% | 76.1% | 87.9% | 92.6% | 94.5% |
| Ever Gave Birth to Two Children | 0.7% | 15.4% | 47.1% | 69.9% | 81.4% | 86.5% |
| Ever Gave Birth to Three Children | 0.0% | 2.9% | 16.0% | 32.5% | 46.7% | 53.6% |
| Ever Gave Birth to Four Children | 0.0% | 0.7% | 5.5% | 14.2% | 21.2% | 26.6% |
| Ever Gave Birth to Five Children | 0.0% | 0.1% | 1.6% | 6.3% | 10.8% | 14.4% |
| Observations | 8,316 | 8,514 | 8,468 | 7,739 | 6,477 | 5,026 |

Higher baseline higher-parity rates than the overall sample (e.g. parity $\geq 4$ by 32: 21.2% vs. 15.9% overall) — room to move up the parity ladder.

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# Appendix: Baseline Parity-Attainment — Kurdish / Arabic Mother Tongue

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | 16.6% | 55.0% | 78.5% | 89.6% | 93.6% | 96.4% |
| Ever Gave Birth to One Child | 5.5% | 41.3% | 69.7% | 84.0% | 90.5% | 93.6% |
| Ever Gave Birth to Two Children | 1.2% | 20.4% | 54.0% | 74.1% | 84.6% | 90.9% |
| Ever Gave Birth to Three Children | 0.1% | 6.0% | 28.8% | 53.6% | 69.4% | 78.7% |
| Ever Gave Birth to Four Children | 0.0% | 1.8% | 14.1% | 33.9% | 48.3% | 60.6% |
| Ever Gave Birth to Five Children | 0.0% | 0.3% | 4.7% | 18.9% | 31.8% | 43.8% |
| Observations | 2,940 | 2,722 | 2,396 | 1,987 | 1,526 | 1,090 |

Substantially higher baseline higher-parity rates (e.g. parity $\geq 4$ by 32: 44.2% vs. 15.9% overall; parity $\geq 5$ by 32: 28.7% vs. 8.0%). Largest scope for upward parity progression.

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# Appendix: Baseline Parity-Attainment — High-Exposure Border Region

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | 12.9% | 50.6% | 75.3% | 87.7% | 92.9% | 95.4% |
| Ever Gave Birth to One Child | 4.6% | 36.5% | 65.5% | 82.0% | 89.6% | 92.4% |
| Ever Gave Birth to Two Children | 0.9% | 17.3% | 48.1% | 69.6% | 81.6% | 88.2% |
| Ever Gave Birth to Three Children | 0.1% | 4.8% | 24.5% | 48.8% | 64.8% | 74.9% |
| Ever Gave Birth to Four Children | 0.0% | 1.4% | 11.9% | 29.7% | 44.2% | 58.2% |
| Ever Gave Birth to Five Children | 0.0% | 0.3% | 3.8% | 15.9% | 28.7% | 41.0% |
| Observations | 3,757 | 3,406 | 2,981 | 2,453 | 1,902 | 1,350 |

Highest baseline higher-parity rates (e.g. parity $\geq 5$ by 32: 28.7% – shared with the Kurdish/Arabic subgroup, since the two cuts overlap heavily in the East/Southeast). Common thread across all three heterogeneity slides: **effects appear where the parity ladder has room left to climb**.

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# Appendix • Robustness

*One slide per check; each returns to slide 10 (Main Results).*

NUTS-1 trends | NUTS-2 trends | Drop post-2011 movers

4-wave decomposition | OLS counterpart

Event study (pre-trends) | Parity-progression hazards

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# Appendix: Robustness — NUTS-1-Region Linear Trends

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | -0.122 | -0.510 | 0.414 | 0.230 | 0.859 | -0.413 |
| | (0.247) | (0.683) | (1.832) | (1.070) | (0.980) | (1.274) |
| | [0.98] | [0.95] | [0.98] | [0.98] | [0.93] | [0.98] |
| Ever Gave Birth to One Child | -0.020 | 0.305 | -0.027 | 1.051 | -1.000 | 0.381 |
| | (0.147) | (0.587) | (2.139) | (1.342) | (1.828) | (1.528) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Gave Birth to Two Children | 0.029 | 0.359 | -0.126 | 0.628 | 1.726 | -0.883 |
| | (0.050) | (0.591) | (0.806) | (2.184) | (3.593) | (1.220) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Gave Birth to Three Children | 0.004 | 0.446** | -0.382 | 0.769 | 2.722 | 1.327 |
| | (0.006) | (0.224) | (0.595) | (1.670) | (2.679) | (1.767) |
| | [1.00] | [0.73] | [1.00] | [1.00] | [0.98] | [1.00] |
| Ever Gave Birth to Four Children | – | – | 0.557 | 1.731** | 1.647 | 4.184* |
| | (–) | (–) | (0.403) | (0.849) | (1.471) | (2.159) |
| | | | [0.93] | [0.72] | [0.97] | [0.73] |
| Ever Gave Birth to Five Children | – | – | 0.304 | 1.601* | 1.108 | 4.020** |
| | (–) | (–) | (0.241) | (0.919) | (0.911) | (1.965) |
| | | | [0.95] | [0.82] | [0.95] | [0.72] |
| Observations | 14,354 | 13,484 | 12,241 | 10,240 | 8,245 | 6,110 |

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# Appendix: Robustness — NUTS-2-Region Linear Trends

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | -0.288 | -1.093 | -0.758 | -1.000 | -0.129 | -0.757 |
| | (0.289) | (0.802) | (1.920) | (1.010) | (1.139) | (1.184) |
| | [0.84] | [0.68] | [0.90] | [0.84] | [0.90] | [0.90] |
| Ever Gave Birth to One Child | -0.107 | -0.264 | -1.124 | 0.022 | -2.168 | -0.729 |
| | (0.182) | (0.771) | (2.515) | (1.436) | (1.822) | (1.575) |
| | [1.00] | [1.00] | [1.00] | [1.00] | [0.95] | [1.00] |
| Ever Gave Birth to Two Children | 0.017 | -0.100 | -0.811 | -1.065 | -0.480 | -2.465 |
| | (0.039) | (0.457) | (1.015) | (2.557) | (3.478) | (1.582) |
| | [1.00] | [1.00] | [0.99] | [1.00] | [1.00] | [0.87] |
| Ever Gave Birth to Three Children | 0.000 | 0.298 | -0.755 | 0.744 | 1.413 | 1.705 |
| | (0.006) | (0.214) | (0.761) | (1.759) | (2.806) | (1.959) |
| | [1.00] | [0.90] | [0.98] | [1.00] | [1.00] | [0.99] |
| Ever Gave Birth to Four Children | – | – | 0.243 | 1.630** | 0.778 | 3.357 |
| | (–) | (–) | (0.411) | (0.831) | (2.428) | (2.636) |
| | | | [1.00] | [0.74] | [1.00] | [0.93] |
| Ever Gave Birth to Five Children | – | – | 0.232 | 1.295 | 1.345 | 3.523 |
| | (–) | (–) | (0.227) | (0.871) | (1.194) | (3.186) |
| | | | [0.98] | [0.89] | [0.96] | [0.96] |
| Observations | 14,354 | 13,484 | 12,241 | 10,240 | 8,245 | 6,110 |

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# Appendix: Robustness — Dropping Post-2011 Out-Migrants

| | Parity $\geq 4$ | | Parity $\geq 5$ | |
|---|---|---|---|---|
| | by 28 | by 32 | by 28 | by 32 |
| Primary sample | 1.905** | 2.246 | 1.392* | 1.414 |
| | (0.809) | (1.415) | (0.803) | (0.912) |
| $N$ | 10,240 | 8,245 | 10,240 | 8,245 |
| Drop post-2011 movers | 3.220*** | 3.914* | 1.668 | 1.399 |
| | (1.101) | (2.059) | (1.177) | (1.026) |
| $N$ | 9,365 | 7,643 | 9,365 | 7,643 |

Concern: women in high-$B_n$ provinces in 2011 may have moved away, attenuating exposure for those who stayed. Movers = 12.3% pooled (7.25% in 2013, 19.53% in 2018). Mover indicator **uncorrelated with** $B_n$ ($\hat{\beta} = 0.002$, $p = 0.73$). Non-mover estimates are **larger and more precise** at the four core higher-parity cells, consistent with cleaner exposure measurement among stayers.

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# Appendix: Robustness — Four-Wave Decomposition Ladder

| Core outcome | (i) Primary 2013/18, full, prov. 2011 | (ii) EM, prov. 2011 | (iii) EM, childhood prov. | (iv) Four-wave 2003–18, EM, childhood prov. |
|---|---|---|---|---|
| Ever Gave Birth to Four Children, age 28 | 1.905** | 1.768** | 1.750* | 1.211 |
| | (0.809) | (0.879) | (0.940) | (1.007) |
| Ever Gave Birth to Four Children, age 32 | 2.246 | 2.260 | 2.098 | 1.198 |
| | (1.415) | (1.375) | (1.543) | (1.517) |
| Ever Gave Birth to Five Children, age 28 | 1.392* | 1.313 | 0.747 | 1.228 |
| | (0.803) | (0.851) | (0.749) | (0.997) |
| Ever Gave Birth to Five Children, age 32 | 1.414 | 1.437 | 1.159 | 1.099 |
| | (0.912) | (0.924) | (1.238) | (1.067) |
| Observations (egb4, age 28) | 10,240 | 9,746 | 9,746 | 19,776 |

Decomposes the gap between primary FS estimates and four-wave EM estimates by changing one feature at a time.

- (i)→(ii): ever-married restriction (small effect).
- (ii)→(iii): replacing 2011 province with childhood province (**attenuates parity 5@28 from 1.31 to 0.75** — measurement error).
- (iii)→(iv): adds 2003/08 waves (further attenuates; less precision).

Reading: **FS spec is preferred baseline**; four-wave EM is a complementary lower-bound check.

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# Appendix: Robustness — OLS Counterpart of the Main 2SLS

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | -0.036 | -0.743 | 0.799 | -1.056 | -0.414 | -0.164 |
| | (0.176) | (0.635) | (1.472) | (1.458) | (0.758) | (0.862) |
| Ever Gave Birth to One Child | -0.013 | -0.259 | 0.655 | -0.185 | -1.190 | -0.478 |
| | (0.131) | (0.418) | (1.327) | (1.561) | (1.145) | (0.975) |
| Ever Gave Birth to Two Children | 0.036 | 0.443 | 0.606 | 0.320 | 0.080 | -0.220 |
| | (0.029) | (0.391) | (0.788) | (1.584) | (2.349) | (1.091) |
| Ever Gave Birth to Three Children | 0.006* | 0.128 | -0.516 | 1.592 | 2.069 | 2.311* |
| | (0.003) | (0.319) | (0.429) | (1.387) | (2.045) | (1.369) |
| Ever Gave Birth to Four Children | – | – | 0.050 | 1.367 | 2.565** | 3.543** |
| | (–) | (–) | (0.263) | (0.909) | (1.130) | (1.543) |
| Ever Gave Birth to Five Children | – | – | 0.020 | 0.808 | 2.888*** | 1.359 |
| | (–) | (–) | (0.353) | (0.718) | (0.742) | (1.301) |
| Observations | 14,354 | 13,484 | 12,241 | 10,240 | 8,245 | 6,110 |

OLS estimates on the same sample / controls / trends as the main 2SLS. **Coefficients are close in magnitude to the 2SLS**: the four core higher-parity cells differ by less than $\sim 1.5$ between OLS and 2SLS, and the sign / pattern is unchanged. The IV machinery is correcting selection-on-province bias but is **not generating the headline pattern**; the same higher-parity signature appears in OLS.

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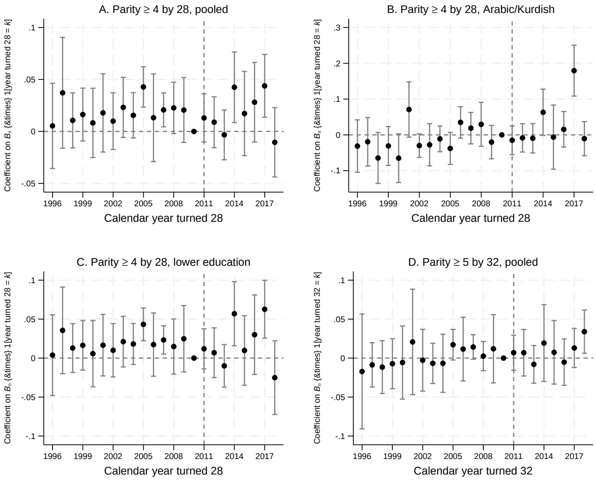
# Appendix: Robustness — Event Study around 2011

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# Appendix: Robustness — Parity-Progression Hazard Decomposition

Discrete-time hazard 2SLS on at-risk person-years for marriage and each birth transition $k \in \{1, \ldots, 5\}$; $R_{j(i,t),t}$ instrumented by $Z_{j(i,t),t} = B_{j(i,t)} \times T_t$:

$$h_{ijt}^{(k)} = \alpha^{(k)} + \beta^{(k)} R_{j(i,t),t} + X_i' \Gamma^{(k)} + \phi^{(k)}_{\mathrm{age}(i,t)} + D_k\, \psi^{(k)}_{\tau(i,t)} + \delta^{(k)}_{j(i,t)} + \mu^{(k)}_t + \theta^{(k)}_{g(j)} t + \varepsilon^{(k)}_{ijt}.$$

| | Marriage | 1st Birth | 2nd Birth | 3rd Birth | 4th Birth | 5th Birth |
|---|---|---|---|---|---|---|
| Refugee ratio $R_{jt}$ | 0.0396 | -0.0568 | -0.0324 | 0.0638 | -0.0056 | 0.0100 |
| | (0.0587) | (0.0426) | (0.1880) | (0.0775) | (0.0677) | (0.0616) |
| | [0.94] | [0.67] | [1.00] | [0.90] | [1.00] | [1.00] |
| Person-years | 142,371 | 166,710 | 54,655 | 63,252 | 33,962 | 15,459 |

**Marriage and first-birth hazards essentially zero**; higher-parity hazards small and noisy. Current-year exposure on person-years cannot localize the cumulative response transition by transition. **Consistent with, not independently confirming,** the cumulative higher-parity story.

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# Appendix • Setting, sample, data

# Appendix: First-Stage $F$ and Coefficient on $\bar{Z}_{ja}$

| | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Panel A: Primary sample (2013/18, full sample, province in 2011)* | | | | | | |
| $\bar{Z}_{ja}$ ($\times 10^{-5}$) | 2.514*** | 2.229*** | 2.269*** | 2.453*** | 2.293*** | 2.310*** |
| | (0.379) | (0.195) | (0.224) | (0.371) | (0.315) | (0.193) |
| First-stage $F$ | 43.9 | 131.3 | 102.4 | 43.7 | 52.9 | 143.7 |
| Observations | 14,354 | 13,484 | 12,241 | 10,240 | 8,245 | 6,110 |
| *Panel B: Four-wave sample (2003/08/13/18, ever-married, childhood province)* | | | | | | |
| $\bar{Z}_{ja}$ ($\times 10^{-5}$) | 2.222*** | 2.359*** | 2.389*** | 2.430*** | 2.251*** | 2.338*** |
| | (0.410) | (0.411) | (0.334) | (0.368) | (0.309) | (0.256) |
| First-stage $F$ | 29.4 | 32.9 | 51.2 | 43.5 | 53.0 | 83.2 |
| Observations | 19,281 | 21,525 | 21,782 | 19,776 | 16,442 | 12,282 |

First-stage $F \in [43, 144]$ across all primary-sample cells; $F \in [29, 83]$ in the four-wave EM sample.
Instrument is **strong** across the board; weak-IV concerns do not bind.

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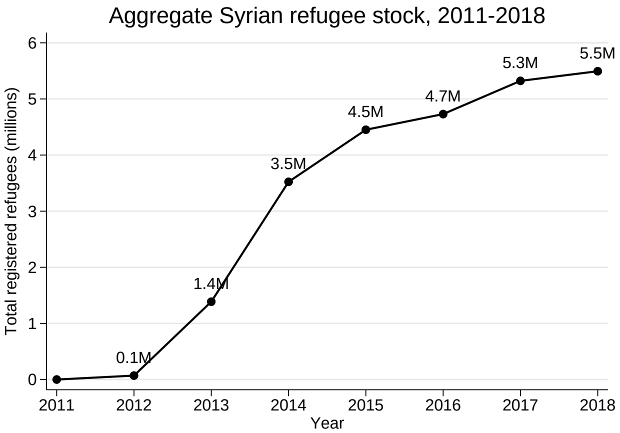
# Appendix: Aggregate Refugee Stock $T_t$ across Host Countries




$T_t$ builds up progressively after 2011 across Turkey, Lebanon, Jordan, and Iraq, rather than as a single step change. The shock activates $B_n$ gradually over the 2011–2018 window.

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# Appendix: TDHS Sample and Sample Restrictions

**Primary sample (full):** TDHS 2013 + 2018.

- Ever- and never-married women aged 15–49 at interview.
- Province assigned by **2011 residence** (pre-shock, predetermined).
- Full controls available (wave, childhood urban/rural, mother's education, mother tongue).

**Four-wave EM (appendix):** TDHS 2003 + 2008 + 2013 + 2018.

- Ever-married only (1998/2003/2008 surveyed only ever-married women).
- Province by **childhood province** (no detailed adult migration history in 2003/08).
- Used as complementary specification; subject to selection + measurement error.

**Why FS leads.** Marriage is itself a treatment-responsive outcome (selection); classical measurement error on province ⇒ attenuation bias toward zero.

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# Appendix: Reconstructing the Pseudo-Panel

**Cross-sectional record (TDHS interview):**

| ID | Birth Yr | Childhood Prov | Move Yr | Dest. Prov | Birth 1 | Birth 2 |
|---|---|---|---|---|---|---|
| 10001 | 1990 | Adana | 2009 | Mardin | 2010 | 2013 |

$\Downarrow$ expanded into woman-by-year panel $\Downarrow$

**Pseudo-panel:**

| ID | Age | Year | Province | Gave Birth 1 | Gave Birth 2 |
|---|---|---|---|---|---|
| 10001 | 12 | 2002 | Adana | 0 | 0 |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| 10001 | 20 | 2010 | Mardin | 1 | 0 |
| 10001 | 23 | 2013 | Istanbul | 1 | 1 |

Outcomes are collapsed to one row per $(i, a)$ for each age cutoff; exposure $\bar{R}_{ja}$ averaged over age 12–$a$.

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## Appendix: Data Sources and Cohort Variation in $\bar{R}_{ja}$

**Province-year ratio.** $R_{j,t}$ = registered Syrians / native population in province $j$, year $t$, from **Ministry of Interior** / **AFAD** administrative records (2012–2018; zero before 2012).

**Cohort variation in $\bar{R}_{ja}$.** A 2013 woman observed at age 28 averages exposure over 1997–2013 (**2 post-shock years**); a 2018 woman at age 28 averages over 2002–2018 (**7 post-shock years**). Same age, same province, *different* $\bar{R}_{ja}$. Crossed with $B_j$, this cross-cohort variation in window timing is what the IV uses on top of FE.

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# Appendix • Complement subgroups

*Panel B of paper heterogeneity tables: the other half of each cut.*

# Appendix: Higher-Educated Women

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Higher education (secondary / tertiary)* | | | | | | |
| Ever Married | 0.018 | 0.214 | -0.886 | -1.412 | -1.633 | 1.927 |
| | (0.053) | (0.589) | (3.014) | (3.269) | (2.920) | (4.802) |
| | [0.99] | [0.99] | [0.99] | [0.99] | [0.99] | [0.99] |
| Ever Birth $\geq 1$ | 0.041 | -0.391 | -0.579 | 0.481 | -9.425* | 0.625 |
| | (0.029) | (0.672) | (2.911) | (3.708) | (4.929) | (5.498) |
| | [0.94] | [0.99] | [1.00] | [1.00] | [0.85] | [1.00] |
| Ever Birth $\geq 2$ | 0.001 | 0.304 | 0.274 | 0.100 | -4.555 | -15.097* |
| | (0.024) | (0.239) | (1.155) | (4.655) | (7.553) | (8.562) |
| | [1.00] | [0.96] | [1.00] | [1.00] | [0.99] | [0.88] |
| Ever Birth $\geq 3$ | – | 0.080 | 0.044 | -3.229** | 7.384 | -18.786** |
| | (–) | (0.061) | (0.806) | (1.266) | (6.983) | (7.432) |
| | | [0.96] | [1.00] | [0.67] | [0.97] | [0.67] |
| Ever Birth $\geq 4$ | – | – | -0.571 | -0.322 | 3.927 | -8.802* |
| | (–) | (–) | (0.497) | (0.370) | (3.189) | (4.863) |
| | | | [0.97] | [0.98] | [0.96] | [0.87] |
| Ever Birth $\geq 5$ | – | – | -0.486 | 0.130 | -0.365 | -2.314 |
| | (–) | (–) | (0.481) | (0.186) | (0.567) | (2.160) |
| | | | [0.97] | [0.99] | [0.99] | [0.97] |
| Observations | 4,993 | 4,352 | 3,512 | 2,607 | 1,897 | 1,213 |

Secondary / tertiary education. Raw point estimates are noisy and not concentrated at higher parities; no cell rejects under raw inference or under RW. Consistent with the lower-education subgroup carrying the higher-parity response.

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# Appendix: Turkish Mother Tongue

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Turkish mother tongue* | | | | | | |
| Ever Married | -0.433** | -1.286 | -0.143 | 0.240 | -1.519 | -2.385 |
| | (0.197) | (0.796) | (1.681) | (1.621) | (1.295) | (2.949) |
| | [0.37] | [0.53] | [0.98] | [0.98] | [0.70] | [0.83] |
| Ever Birth $\geq 1$ | -0.172 | -0.813 | -0.569 | 1.233 | -3.926 | -2.142 |
| | (0.120) | (0.773) | (2.379) | (1.928) | (2.778) | (3.325) |
| | [0.94] | [0.98] | [1.00] | [1.00] | [0.94] | [1.00] |
| Ever Birth $\geq 2$ | -0.003 | 0.062 | -1.223 | 0.191 | -0.779 | -3.663 |
| | (0.024) | (0.747) | (0.906) | (2.764) | (5.461) | (2.398) |
| | [1.00] | [1.00] | [0.94] | [1.00] | [1.00] | [0.92] |
| Ever Birth $\geq 3$ | 0.004 | 0.165 | -0.343 | -1.099 | 2.768 | -0.849 |
| | (0.004) | (0.317) | (0.570) | (1.785) | (4.006) | (3.397) |
| | [0.99] | [1.00] | [1.00] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 4$ | – | – | -0.075 | -0.346 | 2.582** | 1.821 |
| | (–) | (–) | (0.325) | (1.064) | (1.061) | (2.959) |
| | | | [1.00] | [1.00] | [0.70] | [1.00] |
| Ever Birth $\geq 5$ | – | – | 0.261** | -0.043 | 0.026 | 5.116 |
| | (–) | (–) | (0.115) | (0.310) | (1.540) | (3.823) |
| | | | [0.74] | [1.00] | [1.00] | [0.94] |
| Observations | 11,013 | 10,512 | 9,655 | 8,137 | 6,634 | 4,960 |

Turkish mother-tongue women. Point estimates near zero across parities and ages; **no raw or RW rejection at higher parities**. The Kurdish/Arabic subgroup is the one that drives the higher-parity result.

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# Appendix: Rest of Turkey

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| *Rest of Turkey* | | | | | | |
| Ever Married | -0.261 | -1.358 | 1.726 | 0.047 | -2.056 | -1.905** |
| | (0.187) | (1.073) | (1.784) | (1.257) | (1.606) | (0.754) |
| | [0.54] | [0.56] | [0.56] | [0.97] | [0.56] | [0.10] |
| Ever Birth $\geq 1$ | 0.045 | -0.394 | 1.823 | 1.304 | -6.070** | -0.989 |
| | (0.226) | (1.020) | (1.400) | (2.049) | (2.517) | (0.978) |
| | [1.00] | [1.00] | [0.88] | [0.99] | [0.56] | [0.96] |
| Ever Birth $\geq 2$ | -0.036 | -0.595 | -1.202 | 2.665 | -4.895 | -1.143 |
| | (0.043) | (0.616) | (0.967) | (2.110) | (3.171) | (1.641) |
| | [0.98] | [0.97] | [0.89] | [0.89] | [0.80] | [0.99] |
| Ever Birth $\geq 3$ | -0.001 | 0.185 | -0.654 | -0.135 | 0.192 | 0.594 |
| | (0.001) | (0.348) | (0.790) | (3.917) | (3.875) | (2.686) |
| | [1.00] | [1.00] | [0.98] | [1.00] | [1.00] | [1.00] |
| Ever Birth $\geq 4$ | – | – | 0.227 | 1.448 | 1.494 | 6.100** |
| | (–) | (–) | (0.478) | (1.377) | (2.367) | (2.545) |
| | | | [1.00] | [0.95] | [0.99] | [0.56] |
| Ever Birth $\geq 5$ | – | – | 0.580** | 0.892 | 1.485 | 5.437* |
| | (–) | (–) | (0.270) | (1.347) | (1.636) | (2.798) |
| | | | [0.61] | [0.99] | [0.98] | [0.65] |
| Observations | 10,352 | 9,962 | 9,219 | 7,836 | 6,394 | 4,799 |

Provinces outside the East/Southeast high-exposure region. Point estimates small and mostly insignificant; the headline higher-parity response is **absent** away from the heavily-treated border region.

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# Appendix: Placebo with NUTS-1 Trends

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | 0.0004 | 0.0013 | 0.0016* | 0.0012** | 0.0014* | -0.0004 |
| | (0.0005) | (0.0008) | (0.0010) | (0.0005) | (0.0008) | (0.0007) |
| | [0.77] | [0.54] | [0.54] | [0.38] | [0.54] | [0.77] |
| Ever Gave Birth to One Child | 0.0006** | 0.0014 | 0.0012 | 0.0013** | 0.0017** | 0.0002 |
| | (0.0003) | (0.0010) | (0.0013) | (0.0006) | (0.0008) | (0.0007) |
| | [0.72] | [0.89] | [0.98] | [0.69] | [0.72] | [1.00] |
| Ever Gave Birth to Two Children | 0.0001 | -0.0000 | 0.0015* | 0.0025*** | 0.0024** | 0.0006 |
| | (0.0002) | (0.0005) | (0.0008) | (0.0009) | (0.0011) | (0.0012) |
| | [1.00] | [1.00] | [0.76] | [0.54] | [0.72] | [1.00] |
| Ever Gave Birth to Three Children | 0.0000 | 0.0001 | 0.0005 | 0.0002 | -0.0000 | -0.0024 |
| | (0.0000) | (0.0003) | (0.0006) | (0.0007) | (0.0007) | (0.0014) |
| | [0.94] | [1.00] | [0.99] | [1.00] | [1.00] | [0.81] |
| Ever Gave Birth to Four Children | – | – | 0.0005 | 0.0011 | 0.0009 | -0.0012 |
| | (–) | (–) | (0.0004) | (0.0010) | (0.0024) | (0.0028) |
| | | | [0.91] | [0.94] | [1.00] | [1.00] |
| Ever Gave Birth to Five Children | – | – | 0.0000 | 0.0004 | 0.0013 | 0.0009 |
| | (–) | (–) | (0.0002) | (0.0005) | (0.0015) | (0.0029) |
| | | | [1.00] | [0.99] | [0.99] | [1.00] |
| Observations | 12,355 | 11,560 | 10,226 | 8,194 | 6,063 | 4,098 |

Same placebo specification as slide 9 ($Y \sim B_n \times t$ on $t(a, i) < 2012$) but with **NUTS-1-region** linear trends in place of macro-region trends. Coefficients remain near zero; **no RW rejection**.

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# Appendix: Placebo with NUTS-2 Trends

| Outcome | Age 16 | Age 20 | Age 24 | Age 28 | Age 32 | Age 36 |
|---|---|---|---|---|---|---|
| Ever Married | -0.0004 | 0.0017 | -0.0016 | -0.0006 | -0.0011 | -0.0034** |
| | (0.0005) | (0.0011) | (0.0010) | (0.0008) | (0.0013) | (0.0016) |
| | [0.75] | [0.45] | [0.44] | [0.75] | [0.75] | [0.26] |
| Ever Gave Birth to One Child | 0.0010** | 0.0015 | -0.0009 | -0.0009 | -0.0007 | -0.0021 |
| | (0.0004) | (0.0013) | (0.0015) | (0.0009) | (0.0014) | (0.0022) |
| | [0.58] | [0.95] | [0.99] | [0.95] | [0.99] | [0.95] |
| Ever Gave Birth to Two Children | 0.0003 | -0.0010 | 0.0010 | 0.0002 | -0.0021* | -0.0012 |
| | (0.0003) | (0.0010) | (0.0015) | (0.0008) | (0.0011) | (0.0014) |
| | [0.95] | [0.95] | [0.99] | [0.99] | [0.73] | [0.97] |
| Ever Gave Birth to Three Children | 0.0000 | -0.0002 | -0.0004 | -0.0007 | -0.0017 | -0.0067* |
| | (0.0000) | (0.0004) | (0.0009) | (0.0015) | (0.0016) | (0.0035) |
| | [0.99] | [0.99] | [0.99] | [0.99] | [0.95] | [0.69] |
| Ever Gave Birth to Four Children | – | – | 0.0004 | 0.0022 | 0.0028 | -0.0022 |
| | (–) | (–) | (0.0006) | (0.0019) | (0.0041) | (0.0049) |
| | | | [0.99] | [0.95] | [0.99] | [0.99] |
| Ever Gave Birth to Five Children | – | – | -0.0001 | 0.0010 | 0.0037 | 0.0050 |
| | (–) | (–) | (0.0003) | (0.0010) | (0.0024) | (0.0043) |
| | | | [0.99] | [0.95] | [0.83] | [0.94] |
| Observations | 12,355 | 11,560 | 10,226 | 8,194 | 6,063 | 4,098 |

Same placebo specification as slide 9 but with **NUTS-2-region** (subregion) linear trends. Coefficients remain near zero across all cells; **no RW rejection** – pre-trend null is robust to finer trend structures.

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# Appendix • Comparison with Aboulhosn, Aksoy & Özcan

**Aboulhosn, Aksoy & Özcan: *Refugee Immigration and Natives' Fertility*.**

- **Same research question:** how do Syrian refugees affect Turkish native fertility?
- **Different data:** they use the National Survey on Domestic Violence Against Women (**NSDVW**) 2008 and 2014; we use TDHS waves with a pseudo-panel structure.
- **Different design:** cross-sectional DiD on two waves vs. our continuous-DiD on an extended pseudo-panel.
- **Different outcome structure:** we track parity-by-age progression $\mathbb{1}\{\text{parity} \geq p \text{ by age } a\}$; they use contemporaneous 1-year birth/pregnancy indicators.

**AAO headline IV results (Table 1).**

| Outcome | AAO IV |
|---|---|
| Gave birth past year | 0.318** |
| Currently pregnant | 0.248** |
| Number of children | 1.870** |
| Birth & not 1st child | 0.305*** |
| Birth & 1st child | 0.012 |
| Pregnant & $\geq 1$ child | 0.317** |

Reading they offer: *existing mothers*, not childless women, drive the response.

Back Next: replication

# Appendix: AAO Table 1 Replication

**Comparison on AAO's data (NSDVW): paper vs. our replication.**

| Outcome | Paper | Our replication |
|---|---|---|
| Gave birth past year | 0.318** | −0.045 |
| Currently pregnant | 0.248** | 0.612*** |
| Number of children | 1.870** | 2.826*** |
| Birth & not 1st child | 0.305*** | −0.045 |
| Birth & 1st child | 0.012 | 0.000 |
| Pregnant & $\geq 1$ child | 0.317** | 0.574*** |

**Issues we identified in AAO's specification.**

1. **Schooling controls:** asymmetric 2008/2014 recode; 2014 "never attended" $\Rightarrow$ `schooling=0`, 2008 "never attended" $\Rightarrow$ missing (737 women dropped).
2. **Dropping women with missing covariates:** 1,927 women dropped on missing birth-month/year despite valid age (miss count).
3. **Pregnancy definition:** inconsistent coding of missing across samples.
4. **No region time trends:** adding `region`×`year` FE flips the headline.

◂ Back Next: AAO spec on TDHS

# Appendix: AAO Specification on TDHS 2008 + 2013

**Exercise.** Re-run AAO's exact specification (same controls, same IV, same outcomes) on our **TDHS 2008 + 2013** ever-married 15–49 sample ($N = 14{,}621$).

| Outcome | AAO paper IV | TDHS replication IV |
|---|---|---|
| Gave birth past year | $0.318^{**}$ | 0.301 |
| Currently pregnant | $0.248^{**}$ | $-0.465^{**}$ |
| Number of children | $1.870^{**}$ | $2.588^{*}$ |
| Birth & not 1st child | $0.305^{***}$ | $-0.121$ |
| Birth & 1st child | 0.012 | $0.421^{*}$ |
| Pregnant & $\geq 1$ child | $0.317^{**}$ | $-0.223$ |

Back Back to AAO setup